When AI Navigates the Fog of War

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Research Questions

- Can large language models reason coherently about an unfolding geopolitical crisis when armed only with partial, noisy information available at each temporal moment?
- How do LLM strategic reasoning capabilities vary across different domains (economic vs. political vs. military) during real-time geopolitical analysis?
- How do model narratives and predictions evolve as new information becomes available during an ongoing conflict?

Core Idea

The core problem is that evaluating LLM geopolitical reasoning on historical events is fundamentally flawed due to training data leakage - models may have memorized outcomes rather than genuinely reasoning. The key insight is to study an ongoing conflict (2026 Middle East crisis) that occurred entirely after model training cutoffs, creating a natural experiment where models must rely on reasoning rather than memorization. The approach works by constructing 11 temporal nodes with strict information constraints, forcing models to reason from only publicly available information at each moment, thus eliminating hindsight bias and data contamination.

Key Formula

Calibration Consistency Score

$$\text{Calibration Score} = 1 - \text{MAE}, \text{ where } \text{MAE} = \frac{1}{N}\sum_{k=1}^{N} |p_k - y_k|$$

p_k is the probability estimate from model response, y_k ∈ {0,1} is ground truth outcome, N is number of questions. Measures how well model probability estimates align with observed outcomes, with higher values indicating better calibration. Important because it quantifies model reasoning quality under uncertainty.

Key Figure

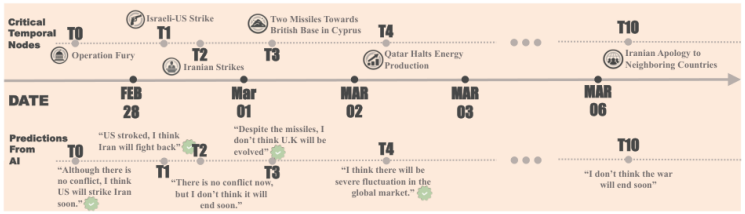

Figure 1 Critical temporal nodes during the early stages of the 2026 Middle East conflict (top) and representative model analyses at selected nodes (bottom). Models exhibit recognizable patterns of strategic reasoning under the fog of war, and their analyses evolve as new information becomes available across the timeline.

This historical thought experiment highlights a broader question about artificial intelligence (AI) reasoning in complex environments. Real-world geopolitical crises involve intertwined dynamics across military strategy, economic incentives, diplomacy, domestic politics, and human perception (Betts, 1978; Jervis, 1976). Effective reasoning in these environments demands the ability to navigate uncertainty, shifting incentives, and partial observability, which are the classic conditions often described as the *fog of war*.

Understanding whether current state-of-the-art (SOTA) Large Language Models (LLMs) exhibit such capabilities is challenging. Retrospective evaluation of historical events is fundamentally confounded by training data leakage (Carlini et al., 2021; Aiyappa et al., 2023; Kang et al., 2024). Major geopolitical events are extensively documented in the vast corpora used to pretrain modern models (Brown et al., 2020; Bender et al., 2021), meaning that models may implicitly encode knowledge of outcomes even when prompted to reason from earlier points in time. As a result, retrospective prediction tasks can blur the distinction between genuine reasoning and latent memorization. Recent work has therefore raised growing concerns that many evaluation benchmarks may inadvertently measure pattern recognition or data leakage rather than true out-of-distribution reasoning ability (Magar and Schwartz, 2022; Sainz et al., 2023; Li, 2023; Yang et al., 2023).

To study how LLMs reason under conditions that more closely resemble real-world uncertainty, we turn to a crisis that unfolded entirely after the training cutoff of current frontier models. *The sudden escalation of the Middle East conflict in late February and early March 2026* (ACLED, 2026) provides a rare opportunity to observe how language models interpret an unfolding geopolitical situation without access to the eventual outcomes. Because the early stages of this crisis occurred outside the training distribution of existing models, it offers *a natural setting in which models must rely on their reasoning processes rather than learned knowledge* as shown in Figure 1.

In this work, we use the early stages of the 2026 Middle East conflict as a temporally grounded case study for analyzing **how LLMs interpret and reason about an unfolding geopolitical crisis under strict information constraints**. We reconstruct a timeline consisting of 11 critical temporal nodes during the early stages of the conflict and formulate 42 event-specific questions and 5 general exploratory questions probing distinct aspects of geopolitical reasoning. The construction of these temporal nodes and questions was human-informed: in addition to reviewing contemporaneous reporting and public records, we incorporated input from individuals with lived experience of the war to improve the ecological validity of the timeline and the relevance of the reasoning probes. These questions span themes including *Initial Outbreak*, *Threshold Crossings*, *Economic Shockwaves*, and *Political Signaling*. At each temporal node, models are provided only with contextual information that would have been publicly available up to that moment and are asked to analyze potential developments and strategic implications.

Rather than treating this setting as a forecasting benchmark, **our objective is to examine how language models behave when confronted with a complex, evolving real-world scenario**. In an ongoing conflict, many of the most important questions are not cleanly exhausted events with timeless binary labels: some concern degrees of escalation, some remain contingent on future developments, and some may be *not yet* rather than definitively

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Figure 1: Shows the temporal timeline of 11 critical nodes (T0-T10) with representative AI predictions, demonstrating how models evolve their strategic reasoning as the conflict unfolds from pre-conflict to active warfare.

Limitations

- Single conflict case study limits generalizability to other geopolitical scenarios or time periods
- Manual extraction of probability estimates from model responses introduces potential researcher bias
- One-week resolution window for 'verifiable' questions is somewhat arbitrary and may not capture longer-term developments
- Ongoing nature of conflict means some ground truth labels may change with longer observation periods